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***Abstract*—Reliable short to medium-term climate forecasts support agricultural planning, water-resource management, and disaster preparedness in the Ethiopian highlands. This paper presents a monthly climate forecasting pipeline for Addis Ababa (2,354 m elevation, 8.98° N, 38.80° E) built from nine years of station observations (January 2017 to July 2025), obtained with the support of the Ethiopian Meteorology Agency. Raw monthly records derived from the daily results for maximum temperature, minimum temperature, precipitation, relative humidity, and duration of sunshine were cleaned, reshaped into a regular monthly time series, and linearly interpolated across the small number of missing values. Seasonal Auto-regressive Integrated Moving Average models with exogenous regressors (SARIMAX) with a twelve-month seasonal period were fitted independently to each variable, with orders selected to reflect the strength of the annual cycle and the persistence of each series. Out-of-sample accuracy was assessed with a twelve-month holdout, yielding mean absolute errors of 0.45°C and 0.37°C for maximum and minimum temperature, 4.74 percentage points for relative humidity, 0.81 h for sunshine duration, and 31.1 mm for precipitation. Twenty-four-month forecasts with 80% confidence intervals were then generated for August 2025 through July 2027. The results reproduce Addis Ababa's bimodal rainy-season structure, project a wetter 2026 relative to 2025, and show markedly higher uncertainty for precipitation than for temperature-related variables, consistent with the intermittent and convective nature of highland rainfall. In addition to the modeling pipeline, we describe a lightweight Flask REST API and a static HTML/JavaScript dashboard that together serve the historical record, the forecasts, and the backtest accuracy metrics for interactive exploration.**

***Index Terms*—Addis Ababa, climate forecasting, Flask API, precipitation, SARIMAX, seasonal time series, sunshine duration, temperature, time-series forecasting, web dashboard.**

Twenty-Four-Month Probabilistic Forecasting of Addis Ababa's Climate Variables Using Seasonal ARIMA Models

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Email: research@example.org Code and data: https://github.com/Bisratolera/weather_forcast_AI

I. Introduction

ADDIS Ababa, the Ethiopian capital, sits at approximately 2,354 m above sea level on the central highland plateau. Its climate is shaped by the seasonal migration of the Inter-Tropical Convergence Zone, producing a pronounced bimodal precipitation regime: a short rainy season ("Belg") in the boreal spring and a longer, more intense rainy season ("Kiremt") from June through September [4], [5]. Inter-annual variability in the onset, duration, and intensity of these rains has direct consequences for agriculture, urban water supply, and flood risk in the metropolitan area.

Short- and medium-term forecasts of temperature, precipitation, humidity, and sunshine duration are therefore of practical value to municipal planners and agricultural stakeholders. Classical seasonal time-series models remain attractive for this purpose because they are interpretable, computationally inexpensive, and, for climatology series with a strong and stable annual cycle, competitive with more complex approaches [2], [3]. This paper documents a reproducible SARIMAX-based pipeline that ingests raw station records supplied with the support of the Ethiopian Meteorology Agency, prepares a monthly time series for Addis Ababa, fits independent seasonal models to five climate variables, validates each model on a held-out year, produces twenty-four-month probabilistic forecasts, and exposes the results through a small backend service and browser-based dashboard.

The remainder of this paper is organized as follows. Section II reviews related work on statistical climate forecasting for East Africa. Section III describes the study area and the data-preparation procedure. Section IV details the forecasting methodology, including model specification and validation strategy. Section V describes the system architecture, covering the Flask backend and the HTML/JavaScript frontend used to serve and visualize the results. Section VI presents the backtest accuracy results and the twenty-four-month forecasts. Section VII discusses the findings and their limitations. Section VIII concludes the paper.

II. Related Work

Statistical characterizations of Ethiopian highland rainfall have long noted its bimodal structure and strong sensitivity to large-scale climate drivers such as the El Niño–Southern Oscillation [4], [5], motivating region-specific rather than globally-tuned forecasting approaches. General-purpose climate model intercomparisons such as CMIP5 [1] provide long-range projections at coarse spatial resolution, but their grid-cell outputs are not well suited to station-level, sub-annual operational planning. Classical Box–Jenkins seasonal ARIMA methods [2], [3] remain a common baseline for short- and medium-term station-level forecasting precisely because they require no auxiliary predictors, are inexpensive to refit as new observations arrive, and give an explicit, interpretable decomposition of trend, seasonality, and noise. The present work follows this tradition, applying independently tuned seasonal ARIMA models to a single, well-characterized highland station rather than attempting a spatially distributed or multivariate reanalysis-driven forecast, and additionally follows current practice in reproducible applied forecasting projects [13] by packaging the trained models behind a documented REST API.

This work uses monthly station observations sourced with the support of the Ethiopian Meteorology Agency and reproducible data-science training resources provided by the Ethiopian Artificial Intelligence Institute (EAII).

III. Data and Study Area

A. Study Area

The station record analyzed in this study corresponds to Addis Ababa (8.98° N, 38.80° E; elevation 2,354 m). The raw source file, provided with the support of the **Ethiopian Meteorology Agency**, contains yearly rows for the period 2017–2026, with one row per combination of year and meteorological element – maximum temperature, minimum temperature, precipitation, relative humidity, sunshine duration, and wind speed – and twelve monthly columns per row.

B. Data Preparation

The raw table was filtered to the Addis Ababa station, reshaped from a wide year-by-month layout into a long format, and averaged across duplicate readings for a given element, year, and month. The result was pivoted into a regular monthly time series indexed by calendar date, spanning January 2017 through May 2026 at the time of writing, with element codes renamed to descriptive column names (e.g., TMPMAX → `temp_max_C`). For the modeling stage described below, the series was restricted to the interval ending July 2025,

TABLE I
PAIRWISE CORRELATION OF MONTHLY SERIES, 2017–2025

	T-max	T-min	Precip.	RH	Sun hrs
T-max	1.00	-0.06	-0.72	-0.80	0.74
T-min	-0.06	1.00	0.55	0.56	-0.62
Precip.	-0.72	0.55	1.00	0.90	-0.92
RH	-0.80	0.56	0.90	1.00	-0.95
Sun hrs	0.74	-0.62	-0.92	-0.95	1.00

TABLE II
SARIMAX ORDERS BY VARIABLE

Variable	(p, d, q)	(P, D, Q, m)
Maximum temperature	$(1, 0, 1)$	$(1, 1, 1, 12)$
Minimum temperature	$(1, 0, 1)$	$(1, 1, 1, 12)$
Precipitation	$(1, 0, 1)$	$(0, 1, 1, 12)$
Relative humidity	$(1, 0, 0)$	$(1, 1, 1, 12)$
Sunshine duration	$(1, 0, 0)$	$(1, 1, 1, 12)$

giving 103 monthly observations per variable, so that a further twenty-four months could be reserved for out-of-sample forecasting.

Wind speed was recorded for only a minority of months (approximately one third of the record) and was excluded from the forecasting stage; the remaining five variables had no missing monthly values over the modeling window and required no interpolation. Table IV reports the historical mean of each modeled variable together with its forecast accuracy.

C. Cross-Variable Relationships

Before modeling, pairwise Pearson correlations among the five monthly series were examined over the 2017–2025 window (Table I). Sunshine duration and relative humidity are strongly and negatively correlated ($r = -0.95$), as expected physically: cloud cover during the rainy season simultaneously raises humidity and blocks sunshine. Precipitation correlates strongly with both relative humidity ($r = 0.90$) and, negatively, with sunshine duration ($r = -0.92$) and maximum temperature ($r = -0.72$), reflecting the cooling and cloud-forming effect of Kiremt-season convection. Minimum temperature is comparatively decoupled from maximum temperature ($r = -0.06$), consistent with the two series being driven by different processes – nighttime radiative cooling versus daytime insolation. These dependencies motivate the joint-modeling extension noted in Section VII but are not enforced in the independent-model pipeline evaluated here.

IV. METHODOLOGY

A. Model Specification

Each of the five climate variables was modeled independently with a Seasonal Auto-regressive Integrated Moving Average model with exogenous regressors, SARIMAX$(p, d, q) \times (P, D, Q)_m$, using a seasonal period of $m = 12$ months to capture Addis Ababa's annual cycle, implemented with the `statsmodels SARIMAX` estimator [7], [11]. In the standard Box–Jenkins notation, the model applied to a (possibly differenced) series y_t is

$$\varphi(B)\,\Phi(B^m)\,(1-B)^d(1-B^m)^D\,y_t = \theta(B)\,\Theta(B^m)\,\varepsilon_t \quad (1)$$

where B is the backshift operator, φ and θ are the non-seasonal auto-regressive and moving-average polynomials of orders p and q, Φ and Θ are their seasonal counterparts of orders P and Q, d and D are the non-seasonal and seasonal differencing orders, and ε_t is white noise. Non-seasonal and seasonal orders were chosen per variable to reflect each series' persistence and the strength of its seasonal differencing requirement: an order of $(1, 0, 1)$ with seasonal order $(1, 1, 1, 12)$ for the two temperature series; $(1, 0, 1)$ with seasonal order $(0, 1, 1, 12)$ for precipitation, whose seasonal amplitude dominates any local trend; and $(1, 0, 0)$ with seasonal order $(1, 1, 1, 12)$ for relative humidity and sunshine duration, as summarized in Table II. All models were estimated with stationarity and invertibility enforcement relaxed to improve numerical convergence on comparatively short series.

B. Validation Strategy

Forecast accuracy was estimated with a fixed-origin holdout: the final twelve months of the 103-month training window were withheld, each model was refit on the remaining months, and twelve-step-ahead forecasts were compared against the withheld observations using mean absolute error (MAE) and root-mean-square error (RMSE). Each model was then refit on the full 103-month series and used to generate a twenty-four-month forecast, from August 2025 through July 2027, with 80% confidence intervals computed from the model's forecast error distribution.

C. Software

The pipeline was implemented in Python [10] using pandas [9] for data preparation, statsmodels [7] (SARIMAX module [11]) for seasonal ARIMA estimation, scikit-learn [8] for error metrics, and Matplotlib [6] for visualization. The data-preparation and modeling stages are implemented as two sequential scripts, so that the monthly time series can be regenerated from raw station records and the forecasts regenerated from the resulting series without manual intervention. Development practices for the pipeline and its packaging followed introductory applied-forecasting material from the Ethiopian Artificial Intelligence Institute's (EAII) Python and machine-learning training curriculum [12] and general community forecasting references [13].

V. SYSTEM ARCHITECTURE AND IMPLEMENTATION

Beyond the offline modeling pipeline, the trained forecasts are served through a small full-stack application consisting of a Python/Flask backend and a static HTML/CSS/JavaScript frontend. Fig. **??** (described in text) summarizes the flow: the two offline scripts write CSV artifacts to a `data/` directory; the Flask application reads those artifacts on request and exposes them as JSON; and the frontend fetches the JSON endpoints and renders the interactive dashboard.

A. Backend: Flask REST API

The backend (`app.py`) is a single-file Flask application, wrapped with `flask_cors` to allow the static frontend to be served from, or fetched by, a different origin during development. Three CSV artifacts produced by the offline pipeline are treated as the source of truth:

- `monthly_timeseries.csv` – the cleaned historical monthly record;
- `forecast_24months.csv` – the twenty-four-month SARIMAX forecast with confidence bounds;
- `model_accuracy_summary.csv` – the twelve-month holdout MAE/RMSE per variable.

On each request, the relevant CSV is loaded with pandas, dates are parsed and sorted, a calendar-month-to-season lookup (`SEASONS`) tags every row with "Bega," "Belg," or "Kiremt," and a helper (`_clean`) replaces any `NaN`/`NaT` values with `None` so that the payload serializes to valid JSON – `NaN` is not valid JSON and would otherwise break `JSON.parse` on the client. A representative excerpt of the request-handling logic is shown in Listing 1.

Listing 1. Core request-handling pattern used by every backend route.

```
def _clean(records):
    """Replace NaN/NaT with None so the payload is
       valid JSON."""
    return [
        {k: (None if (isinstance(v, float)
             and np.isnan(v)) else v)
         for k, v in row.items()}
        for row in records
    ]

def load_historical():
    df = pd.read_csv(HISTORICAL_PATH, parse_dates=["
        date"])
    df = df.sort_values("date")
    df["date"] = df["date"].dt.strftime("%Y-%m-%d")
    df["season"] = df["date"].apply(
        lambda d: SEASONS[int(d[5:7])])
    return _clean(df.to_dict(orient="records"))

@app.get("/api/historical")
def historical():
    return jsonify(load_historical())
```

Table III lists the resulting endpoints. `/api/summary` and `/api/combined` are convenience aggregation routes: rather than requiring the frontend to issue three separate requests and stitch the results together, the backend performs that join server-side and returns a single payload tailored to the dashboard's header and chart components respectively. The remaining two routes (`/` and `/<path:filename>`) serve `index.html` and its static assets directly from the Flask process via `send_from_directory`, so the whole application can be run from a single `python app.py` command without a separate web server.

Because every route re-reads the CSV artifacts rather than caching them in memory, refreshing the forecast (Section VII) only requires overwriting the three CSV files with the output of a re-run pipeline; no backend code changes or restarts of application state are needed, and the API immediately reflects the updated forecast origin on the next request.

TABLE III
BACKEND API ENDPOINTS

Endpoint	Purpose
`GET /api/health`	Liveness check
`GET /api/location`	Station metadata (lat/lon/elev.)
`GET /api/historical`	Full observed monthly series
`GET /api/forecast`	24-month SARIMAX forecast
`GET /api/accuracy`	Holdout MAE/RMSE per variable
`GET /api/variables`	Variable labels and units
`GET /api/summary`	Latest-observed + next-forecast digest
`GET /api/combined`	Historical and forecast stitched per variable
`GET /`	Serves the frontend (`index.html`)

B. Frontend: Interactive Dashboard

The frontend is a static `index.html` page (with companion `style.css` and `script.js`, not reproduced in full here) that consumes the JSON endpoints in Table III and renders five coordinated views:

1) **Now / Next panel** – two side-by-side cards, populated from `/api/summary`, showing the last observed month and the first forecast month for every variable, separated by a "forecast begins" divider;
2) **Season strip** – a horizontal strip spanning the observed-plus-forecast timeline, colored by the Bega/Belg/Kiremt season lookup computed on the backend, giving an at-a-glance sense of Addis Ababa's three-season calendar rather than the conventional four;
3) **Variable chart** – an inline SVG chart (drawn directly into an `<svg>` element rather than an external charting library) with tabs for each of the five variables, plotting the observed series, the projected series, and the shaded 80% confidence band, backed by `/api/combined`;
4) **Accuracy grid** – a set of small comparison bars, one per variable, built from `/api/accuracy`, communicating backtest MAE/RMSE relative to each variable's historical mean;
5) **About section** – three short cards explaining the data source, the SARIMAX method, and the meaning of the confidence interval, aimed at non-technical readers of the dashboard.

Typography and layout use the Fraunces, Space Grotesk, and JetBrains Mono type families loaded from Google Fonts, and the page includes an Amharic rendering of "Addis Ababa" ( ) in the masthead alongside the station's coordinates and elevation. No client-side framework is used; DOM population and the SVG chart are written in vanilla JavaScript against the JSON payloads described in Section V-A, keeping the deployment footprint to a single Flask process plus static files.

VI. RESULTS

A. Backtest Accuracy

Table IV summarizes twelve-month holdout accuracy for each variable alongside its historical mean, giving a sense of scale for the error magnitudes. Both temperature series were forecast to within half a degree Celsius on average (MAE of 0.45°C for maximum temperature and 0.37°C for minimum temperature), against a historical range spanning roughly 18°C

TABLE IV
TWELVE-MONTH HOLDOUT ACCURACY BY VARIABLE

Variable	Unit	MAE	RMSE	Hist. Mean
Maximum temperature	°C	0.45	0.52	24.08
Minimum temperature	°C	0.37	0.48	11.49
Precipitation	mm/mo.	31.05	41.10	104.30
Relative humidity	%	4.74	6.07	58.52
Sunshine duration	h/day	0.81	1.02	6.86

TABLE V
ILLUSTRATIVE EXCERPT OF THE 24-MONTH FORECAST

Month	T-max (°C)	T-min (°C)	Precip. (mm)	RH (%)
Aug 2025	21.1	12.8	274.1	77.4
Sep 2025	21.9	12.3	196.3	73.1
Oct 2025	23.3	11.8	30.8	56.8
Nov 2025	23.5	10.1	9.1	52.9
Feb 2026†	25.4	10.8	0.9	48.0
Jul 2026†	22.2	13.0	261.4	78.5

to 28°C for daily maxima. Relative humidity and sunshine duration were forecast with MAEs of 4.74 percentage points and 0.81 h respectively, both small relative to their seasonal swings of roughly 45 percentage points and 7 h. Precipitation was the most difficult series to forecast in absolute terms, with an MAE of 31.05 mm and RMSE of 41.10 mm against a historical monthly mean of 104.3 mm and a maximum of nearly 430 mm; this reflects the intrinsically noisy, event-driven nature of highland convective rainfall rather than a deficiency specific to this model.

B. Twenty-Four-Month Forecasts

Fig. 1 shows the full 2017–2025 historical series together with the twenty-four-month forecast and its 80% interval for each of the five variables. The forecasts reproduce Addis Ababa's characteristic annual cycle: maximum temperature peaks near 25.5°C around February–March and dips to approximately 21.1°C during the Kiremt rains, while minimum temperature follows an inverse pattern, falling to roughly 9.4°C during the dry season and rising to about 13.5°C in the wet months. Relative humidity and precipitation move together, both peaking sharply during July and August of each forecast year, while sunshine duration is lowest during the same wet months and highest during the November–February dry season.

Table V lists a short excerpt of the forecast for illustration; entries marked with † fall in the second forecast year (2026) and are included to show the model's projected year-over-year seasonal repetition. Aggregating the forecast by calendar year, total precipitation is projected at approximately 1,174 mm for 2026 – close to the 2017–2024 historical average of roughly 1,265 mm per year and markedly wetter than the partial 2025 total of 620 mm observed through July, which fell during a comparatively dry Belg and early Kiremt season.

VII. DISCUSSION AND LIMITATIONS

The temperature-related variables are forecast with tight, temperature-like confidence intervals throughout the two-year

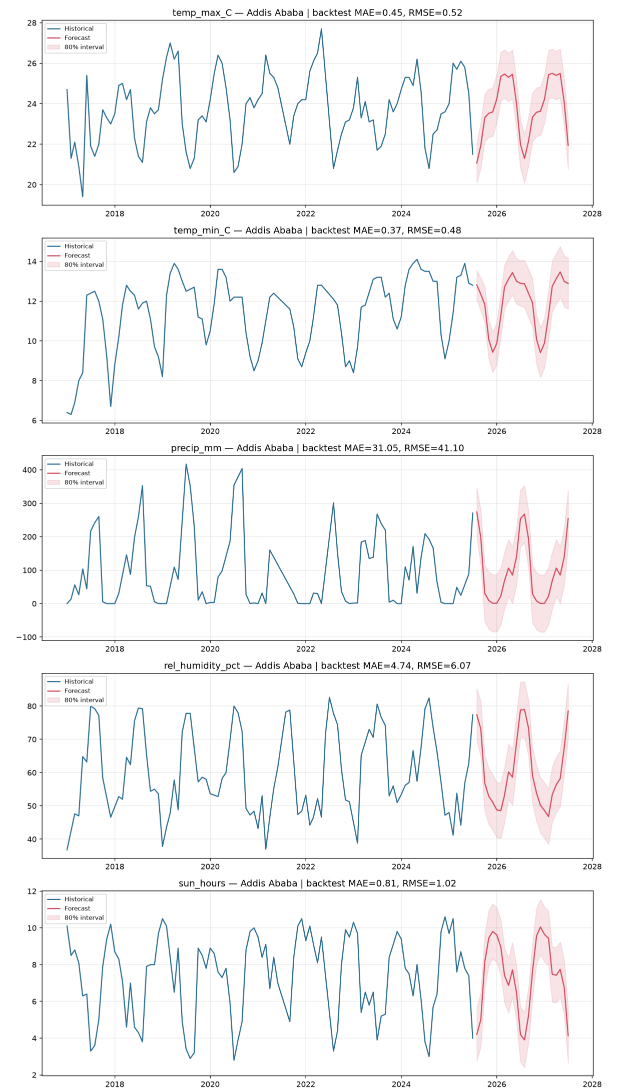

Fig. 1. Historical monthly observations 2017–2025 (blue) and 24-month SARIMAX forecasts with 80% confidence intervals 2025–2027 (red) for maximum temperature, minimum temperature, precipitation, relative humidity, and sunshine duration at Addis Ababa. Rendered interactively in the dashboard described in Section V-B.

horizon, consistent with the strong, stable seasonality and comparatively low year-to-year variability of Addis Ababa's diurnal temperature range at this elevation. Precipitation intervals, by contrast, widen considerably by the second forecast year and in places extend to physically implausible negative values at the lower bound; this is an artifact of using a symmetric, approximately Gaussian interval around a bounded, right-skewed quantity, and any operational use of the precipitation forecast should treat the lower bound as effectively zero rather than as a literal negative rainfall projection. The dashboard described in Section V-B currently renders this band as-is and would benefit from clipping it at zero for precipitation specifically.

A second limitation concerns data currency: the forecasts reported here were generated from the monthly series ending July 2025, whereas the underlying station record – sourced with the support of the Ethiopian Meteorology Agency – has since been extended through May 2026. Re-running the two offline pipeline scripts on the extended series and overwriting the CSV artifacts consumed by the Flask backend (Section V-A) would shift the forecast origin forward and is expected to sharpen near-term intervals as ten additional months of ground truth become available; the system is designed to support this refresh without any application-code changes. Third, each variable is modeled independently; cross-variable dependence (for example, the physical coupling between humidity, cloud cover, and sunshine duration documented in Table I) is not explicitly enforced, though the historical correlation between these series is implicitly captured to the extent that they share a common seasonal cycle. Finally, wind speed could not be modeled owing to substantial missingness in the source record and is a candidate for inclusion, and for a dedicated dashboard tab, once a more complete record becomes available.

VIII. Conclusion

This paper presented a seasonal ARIMA forecasting pipeline for five monthly climate variables at Addis Ababa, validated against a twelve-month holdout and used to generate twenty-four-month forecasts with associated uncertainty bounds, and described the Flask backend and browser-based dashboard used to serve those results. Backtest errors were small relative to each variable's historical range for temperature, humidity, and sunshine duration, and the resulting forecasts reproduce the station's bimodal rainy-season structure. Precipitation remains the hardest-to-forecast variable in absolute terms, and its confidence intervals should be interpreted with the caveats discussed above. Future work includes refreshing the forecast origin as new observations accumulate, exploring variable-specific transformations to better bound the precipitation interval, evaluating whether jointly modeling correlated variables improves on the independent-model approach used here, and extending the dashboard with a wind-speed tab and server-side interval clipping for precipitation.

Acknowledgment

The authors thank the Ethiopian Meteorology Agency for providing and maintaining the station-level data-collection infrastructure that produced the underlying Addis Ababa monthly record used in this study, and for granting access to the raw data used in the forecasting pipeline. The authors also thank the Ethiopian Artificial Intelligence Institute (EAII) for the Python and applied-machine-learning training resources that informed the pipeline's implementation and packaging.

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